SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-019: Student Profile Editing

Field	Details
Change Request ID	CR-019
Change Request Title	Student Profile Editing
Student Name	K.Varshika
Student ID	JF/ICT/24/10
Implementation Status	Completed – subject to verification
Date	29 October 2026

1. Change Request Summary

1.1 Business Requirement

Students needed the ability to update their own contact information without contacting the administrator. The change was required to allow students to edit their email address and phone number through the Student Dashboard.

1.2 Existing Behaviour

Before implementation:

- Students could view their profile information through the Student Dashboard.
- Email address and phone number were displayed as read-only information.
- Students could not directly update their own contact details.
- Administrators managed student information through the administrator dashboard.
- The existing student authentication system used JWT tokens.

1.3 Requested Change

A profile editing facility was required in the Student Dashboard.

The system must:

- Allow students to edit their email address and phone number.
- Keep registration number and student name read-only.
- Validate the email address and phone number.
- Accept only valid Gmail addresses and 10-digit phone numbers.
- Save updated information to the database.
- Display success and error messages.

- Provide Save Changes and Cancel options.
- Allow only authenticated students to update their own profiles.
- Preserve existing administrator student management functionality.

2. Implementation

2.1 Implementation Summary

The CR-019 requirements were implemented in the Student Dashboard. An Edit Profile button was added to allow students to update their email address and phone number.

React state was used to manage the edit form, input values, validation errors, and saving status. Client-side validation checks the email address and phone number before submitting the form.

The frontend sends a PUT request to the backend endpoint `/api/students/me`. The backend identifies the logged-in student using the authenticated user's ID and updates the corresponding student record.

The implementation also provides Save Changes, Cancel, success messages, and error handling.

2.2 Affected Files

File	Change	Purpose
`frontend/src/pages/StudentDashboardPage.jsx`	Modified	Added profile editing form, validation, Save Changes, Cancel, and status messages
`frontend/src/styles/studentDashboard.css`	Modified	Added styling for the profile editing form, buttons, and messages
`frontend/src/services/api.js`	Existing functionality reused	Sends authenticated API requests using the stored JWT token
`backend/src/routes/studentRoutes.js`	Modified	Added the authenticated student profile update route
`backend/src/controllers/studentController.js`	Modified	Validates profile updates and handles the update request

`backend/src/models/studentModel.js`	Modified / to be verified	Updates email and phone number in the database

2.3 Technical Changes

Profile Editing Interface

An Edit Profile button was added to the Student Dashboard. When selected, the email and phone number fields become editable.

The registration number, first name, and last name remain unchanged and cannot be edited through this form.

Input Validation

The frontend validates the entered information before sending the request.

- Email address must be a valid Gmail address.
- Phone number must contain exactly 10 digits.
- Empty fields are rejected.
- Validation errors are displayed beside the relevant input fields.

Backend validation is also required to prevent invalid data from being saved by bypassing the frontend.

Backend API Integration

The following endpoint is used to update the logged-in student's profile:

```
PUT /api/students/me
```

The request sends the email address and phone number in JSON format. The API uses the authenticated user's ID to identify the correct student record.

Authentication and Authorization

The profile update route requires authentication and the `STUDENT` role. The student ID is not accepted from the frontend to determine which record should be updated.

This prevents students from using the profile editing endpoint to update another student's contact details.

Database Update

The backend updates the email and phone number in the existing `students` table. Duplicate email addresses must be checked before updating the record.

User Feedback

The interface displays a success message after a successful update and an error message when the request fails. The Cancel button discards unsaved form changes.

2.4 Implementation Logic

1. The student logs in to the system.
2. The Student Dashboard requests the student's profile using `GET /api/students/me`.
3. The existing profile information is displayed.
4. The student selects Edit Profile.
5. The system displays editable email and phone number fields.
6. The student enters the updated information.
7. The frontend validates the entered values.
8. If the inputs are valid, the frontend sends a `PUT /api/students/me` request.
9. The backend authenticates the request and verifies the student role.
10. The backend validates the input and checks for duplicate email addresses.
11. The database is updated with the new contact information.
12. The updated profile is returned and displayed to the student.
13. A success or error message is displayed according to the result.

2.5 Error and Data Handling

The implementation includes input validation and error handling.

- Empty email and phone number fields are rejected.
- Invalid Gmail addresses are rejected.
- Phone numbers containing other than 10 digits are rejected.
- Authentication failures redirect the student to the login page.
- Duplicate email addresses must be rejected by the backend.
- API errors are displayed to the user.
- The Cancel option restores the previously saved values.
- Registration number and student name remain unchanged by the profile editing operation.

3. Testing & Verification

3.1 Testing Approach

The implementation should be verified through manual frontend and backend testing.

Testing Types:

- Functional testing
- Positive testing
- Validation testing
- UI testing
- Authentication and authorization testing
- Database update testing
- Error handling testing
- Regression testing

Environment: Development

Browser: Google Chrome

Test Role: Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Student Dashboard	Student profile is displayed	To be recorded after testing	Pending
TC-02	Click Edit Profile	Email and phone fields become editable	To be recorded after testing	Pending
TC-03	Enter a valid Gmail address	Email validation succeeds	To be recorded after testing	Pending
TC-04	Enter an invalid email address	Validation error is displayed	To be recorded after testing	Pending
TC-05	Enter a phone number with exactly 10 digits	Phone validation succeeds	To be recorded after testing	Pending
TC-06	Enter an invalid phone number	Validation error is displayed	To be recorded after testing	Pending
TC-07	Save valid profile changes	Updated information is saved and displayed	To be recorded after testing	Pending
TC-08	Click Cancel	Unsaved changes are discarded	To be recorded after testing	Pending
TC-09	Enter an email already used by another student	Backend rejects the duplicate email	To be recorded after testing	Pending

TC-10	Attempt profile update without authentication	Request is rejected	To be recorded after testing	Pending
TC-11	Attempt to update another student's profile through this endpoint	Only the logged-in student's profile can be updated	To be recorded after testing	Pending
TC-12	Verify existing administrator student management	Existing administrator operations continue to work	To be recorded after testing	Pending

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pending	0before testing
Pass Rate	100%

Overall Test Result: To be recorded after testing.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Students can edit their own email address and phone number.	Profile editing test	To be verified
AC-002	Registration number and student name remain read-only.	UI verification	To be verified
AC-003	Invalid email addresses are rejected.	Email validation test	To be verified
AC-004	Phone numbers must contain exactly 10 digits.	Phone validation test	To be verified
AC-005	Valid changes are saved to the database.	Database update test	To be verified
AC-006	Success and error messages are displayed.	UI and error handling tests	To be verified
AC-007	Cancel restores the previously saved values.	Cancel functionality test	To be verified
AC-008	Only authenticated students can update their own profiles.	Authentication and authorization tests	To be verified
AC-009	Existing administrator student management continues to work.	Regression testing	To be verified

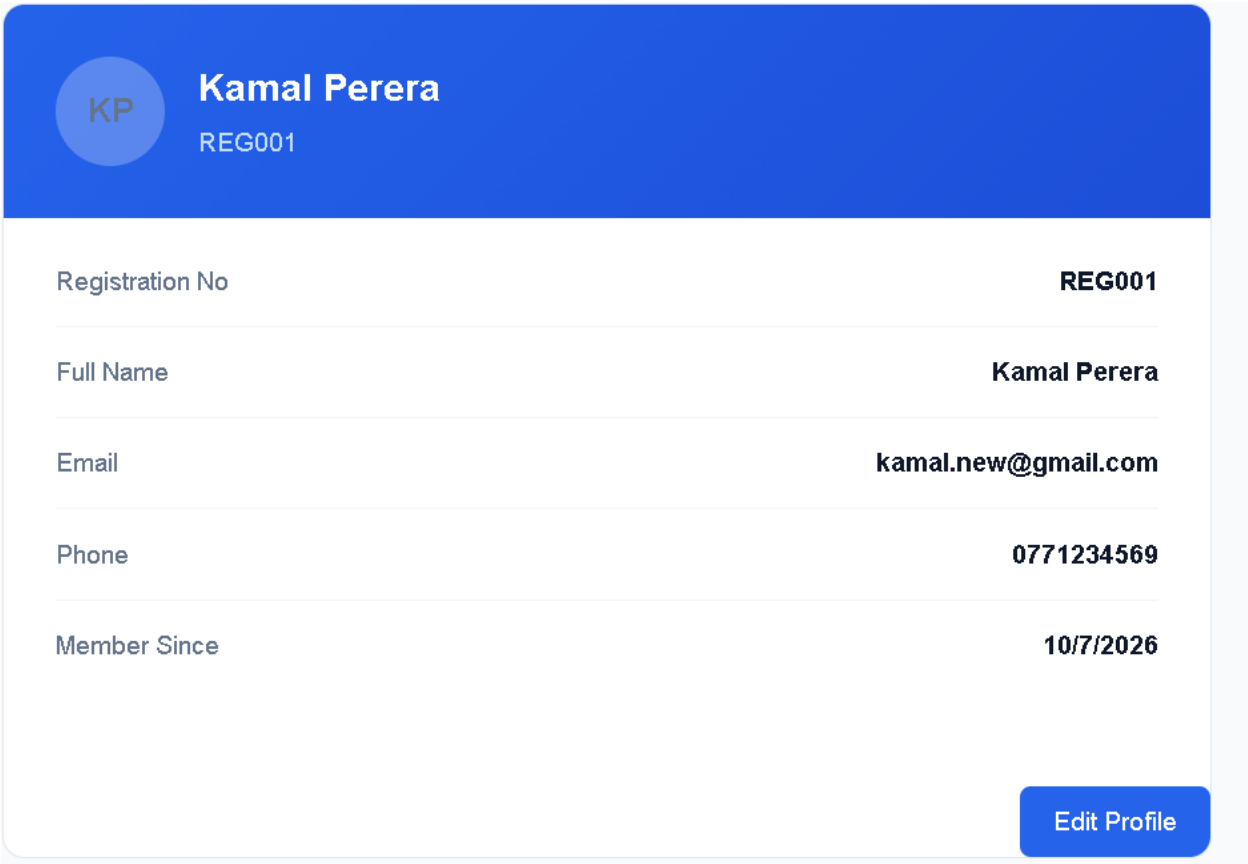
5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Student Profile Editing

Related: AC-001, AC-002, AC-003, AC-004

Figure 1: Student Dashboard showing the Edit Profile form with email and phone number fields.



Evidence 02 – Updated Profile

Related: AC-005, AC-006, AC-007

Figure 2: Student Dashboard showing the updated contact information and success message after saving the changes.

My Profile

KP

Kamal Perera

REG001

Email Address

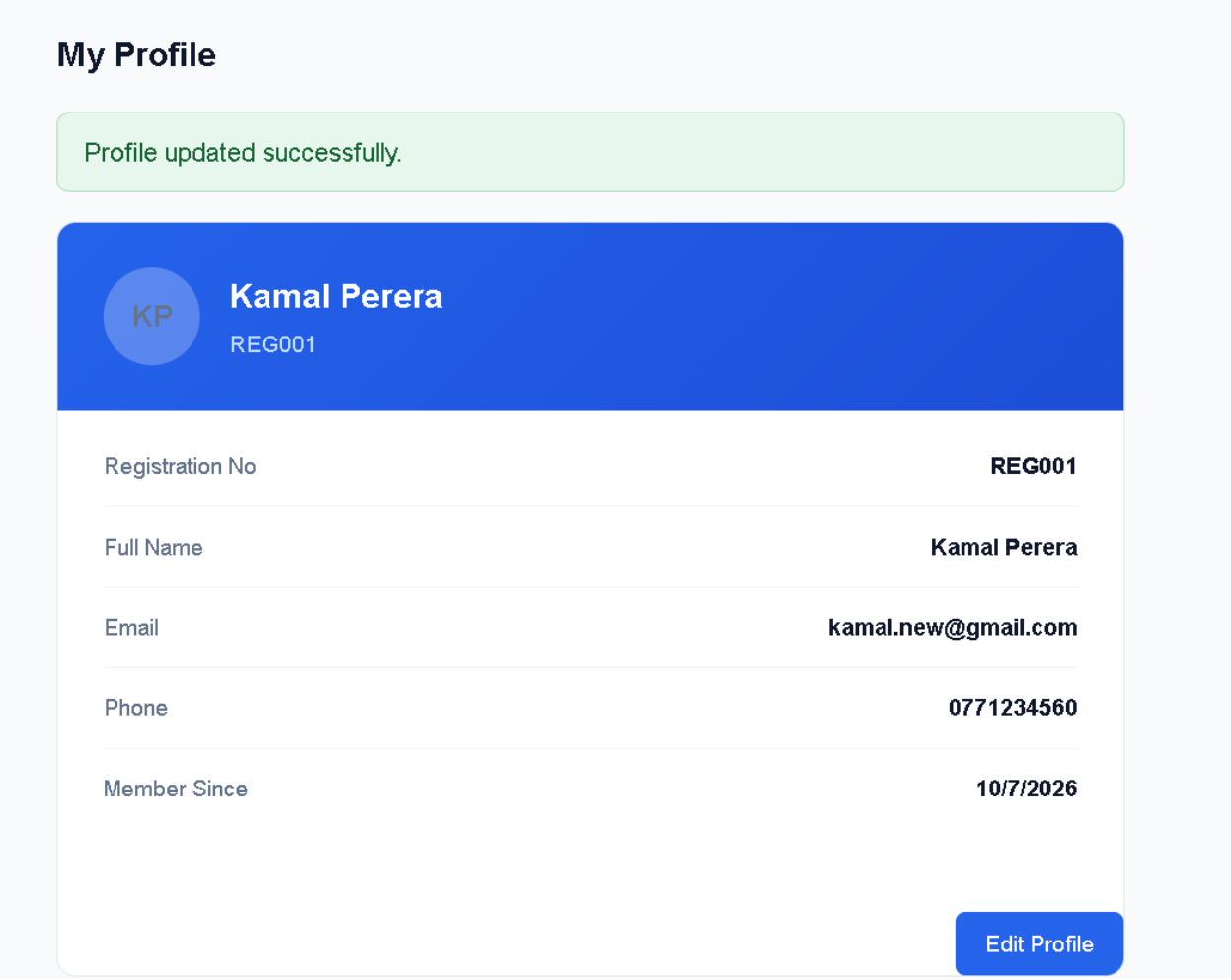
kamal.new@gmail.com

Phone Number

0771234569

Save Changes

Cancel



6. Final Result

Item	Result
Requirements	To be confirmed after verification
Implementation	Implemented in the supplied frontend code; backend integration to be verified
Testing	Pending
Acceptance Criteria	Pending verification
Authentication and Authorization	To be verified
Existing Administrator CRUD	To be verified
Final Status	Pending testing

6.1 Final Summary

CR-019 introduces a profile editing facility for the Student Dashboard. The feature allows authenticated students to update their own email address and phone number while keeping their registration number and name unchanged.

The frontend provides input validation, Save Changes and Cancel options, and success and error messages. The backend endpoint `PUT /api/students/me` is intended to identify the logged-in student through authentication and update the corresponding database record.

The implementation must be tested to confirm successful database updates, validation, authentication, authorization, and compatibility with the existing administrator student management functionality before the change request can be marked as fully completed.

6.2 Issues / Limitations

The following items must be verified before final acceptance:

- The backend profile update route and controller are correctly connected.
- The model updates the correct student record.
- Duplicate email validation works correctly.
- The frontend and backend validation rules are consistent.
- All defined test cases pass.

If all tests pass and no issues remain, this section can be updated to: **No significant issues identified.**

Student Name: K.Varshika

Student ID:JF/ICT/24/10

Date: 29 October 2026

Signature: K.Varshika

~ END OF REPORT ~